2026 OOS evaluation and conclusions

Per-coin P&L and joint basket alpha/betas, with confidence intervals, for all nine frozen strategies. Cash-excess performance at 3% nominal.

Executive Summary

The main finding is failure to generalize across periods

Follow-up orientation through 30 September 2026, not an input to the original development selection. All four development-selected Track A strategies lost in 2024-2025; their raw Sharpes ranged from **-0.64 to -0.22 (365-day)**, or **-0.53 to -0.18 (252-day)**. All five validation-informed Track B strategies lost compounded net return in January-September 2026. B06 still had slightly positive arithmetic raw Sharpe, **0.05 / 0.04 (365 / 252)**; a compounded loss does not require negative Sharpe.

This pattern is **consistent with overfitting or weak transfer, not proof of one cause**: baskets, settings and cadence differ, and Track B also adds long/cash trend. A03's **+31.38%** in 2026 and raw Sharpe **2.32 / 1.93** do not erase its **-22.31%** validation loss. No new winner or reliable all-period strategy is established.

Sharpe convention: tables show **365 / 252** on the **same UTC daily series including weekends**. The alternative is S252 = S365 × √(252/365) = 0.8309097 × S365; it does not drop weekends. An ≈ marks a 252-day value available only from a rounded 365-day source. All primary 365-day results, inference, rankings, CAGR, alpha and 0.03/365 cash accrual are unchanged. Full-precision convention comparison.

Separate fixed all-22 studies did not establish persistence. Development / validation / Jan-Sep 2026 net returns were **+5.61% / -17.43% / -20.94%** for momentum and **-33.59% / -15.66% / +9.91%** for reversal. Neither beat the 3% nominal cash benchmark in all three periods. These remain separate fixed studies, not a new direction selection or fresh independent confirmation. Study results, separate tests and chronology disclosures.

Nine-case and wider-cohort results

One of the nine frozen strategy-basket pairs met the approved joint criteria in January-September 2026; the wider validation screen did not show reliable transfer. A03 returned **+31.38% net**, with **2.32 net Sharpe (365-day), or 1.93 (252-day)**, **7.28% maximum drawdown** and **Holm-9 p = 0.0216**. Its earlier validation return was **-22.31%**, so this is not stable profitability across every period.

The exploratory cohort was substantially weaker: only **17.72%** of the **69,108 validation survivors** gained after 20 bps costs, versus **27.16%** of controls. The stricter **39,332 both-positive** subgroup still had **-10.95% median net return**. Settings and basket composition explain part of the observed gap; uncertainty does not establish that screening is harmful.

No new 2026 winner, basket or portfolio was selected. This is a **273-day partial-year OOS evaluation**. Short availability, funding, order books and realistic fills remain unverified.

Separate fixed all-22 studies did not establish persistent performance

These are **two separately approved simplicity studies**, responding to the proposal to remove basket/parameter selection. Each fixes one 28-day cross-sectional rank-demean rule on all 22 eligible economic slots, with Monday 00:00 UTC rebalancing and no buffer. Momentum buys relative winners and shorts losers; the independent reversal study buys relative losers and shorts winners. No direction is selected, no tests are pooled, and neither changes the nine frozen strategies.

Momentum gained modestly in development but trailed cash; validation and partial-year 2026 lost. **Reversal** lost in development and validation; a positive partial 2026 does not satisfy its all-period condition. The pattern across periods—not the best isolated result—is the finding.

Standalone momentum: unchanged 20 bps / lag1 primary

Period	Net return: raw / vs cash	Raw Sharpe 365 / 252	Cash-excess Sharpe 365 / 252	Hourly max DD	Annual full-L1 turnover
2022–2023 development	Raw 5.61% Cash -0.54%	0.23 / 0.20	0.09 / 0.08	29.53%	33.58×
2024–2025 validation	Raw -17.43% Cash -22.24%	-0.38 / -0.31	-0.53 / -0.44	26.56%	34.91×
Jan–Sep 2026 (273 days)	Raw -20.94% Cash -22.70%	-1.70 / -1.41	-1.87 / -1.55	22.06%	39.29×

Standalone reversal: unchanged 20 bps / lag1 primary

Period	Net return: raw / vs cash	Raw Sharpe 365 / 252	Cash-excess Sharpe 365 / 252	Hourly max DD	Annual full-L1 turnover
2022–2023 development	Raw -33.59% Cash -37.45%	-0.87 / -0.72	-1.01 / -0.84	43.69%	33.58×
2024–2025 validation	Raw -15.66% Cash -20.58%	-0.32 / -0.27	-0.47 / -0.39	31.57%	34.91×
Jan–Sep 2026 (273 days)	Raw 9.91% Cash 7.48%	0.81 / 0.67	0.64 / 0.53	11.10%	39.29×

January–September 2026 (273 days): additional primary risk and growth

Study	Gross return	CAGR: raw / vs cash	Net volatility √365	Daily DD: raw / vs cash	Hourly DD episode
Momentum	-16.15%	Raw -26.96% Cash -29.12%	17.58%	Raw 21.40% Cash 23.13%	Peak 2026-01-01 Trough 2026-09-28 Not recovered; censored 272.8 days underwater
Reversal	16.57%	Raw 13.47% Cash 10.12%	17.54%	Raw 10.77% Cash 11.97%	Peak 2026-05-20 Trough 2026-08-19 Not recovered; censored 133.8 days underwater

Reading the paired values. Raw means strategy return **net of transaction fees**, not gross return. Cash return is W(strategy)/W(cash) − 1; cash CAGR is relative annualized growth, while excess Sharpe subtracts 0.03/365 per daily return. A positive arithmetic excess Sharpe can coexist with negative relative compounded wealth, as in momentum development. The hypothetical **3% nominal ACT/365F cash proxy is not credited** to the strategy. Both Sharpe scales use the same UTC daily observations including weekends. CAGR, volatility, alpha, cash and inference remain calendar-365. Turnover is full-L1 traded-weight multiples including entry, forced events and liquidation. The nine-month CAGR is an extrapolation, not a realized annual return.

January–September 2026 (273 days): uncertainty remains material

Each study has its **own three-period** positive cash-mean test family. No period in either independent study passed its own Holm3 threshold. The intervals below are **marginal and conditional**, not adjusted for earlier research or simultaneous search-wide guarantees.

January–September 2026 (273 days): cash-excess inference

Study	HAC27 t	One-sided p	Own Holm3 p	Return vs cash 95% CI	Excess Sharpe 95% CI: 365 / 252
Momentum	-1.991	0.9813	1.0000	[-41.19%, -1.71%]	365 [-3.65, -0.07] 252 [-3.03, -0.05]
Reversal	0.705	0.2195	0.6584	[-14.29%, 38.17%]	365 [-1.27, 2.37] 252 [-1.06, 1.97]

January–September 2026 (273 days): descriptive BTC sensitivity, marginal HAC7 intervals

Study	Raw alpha/year [95% CI]	Cash-adjusted alpha/year [95% CI]	BTC beta [95% CI]
Momentum	-29.52% [-63.90%, 4.85%]	-32.76% [-67.14%, 1.62%]	-0.079 [-0.124, -0.035]
Reversal	13.85% [-19.84%, 47.55%]	11.07% [-22.62%, 44.77%]	0.074 [0.028, 0.119]

The fixed economic, risk and statistical conditions did not all hold

Each study required **positive cash-relative net return in all three periods**, hourly drawdown **≤40% in all three**, and all three positive-cash-mean tests estimable, positive-t and own-Holm3 p≤0.05. Momentum failed economics and statistical support but passed the drawdown guardrail. Reversal failed all three dimensions; development hourly DD was **43.69%**. Neither is deployment-ready.

Inference: Bartlett HAC27 with n/(n–1); 10,000 circular 28-day bootstrap draws per period, PCG64 seed 20261020 reset per period. The positive-mean test centers the null and uses add-one bootstrap-t p-values. Mean intervals are studentized; return/growth/Sharpe intervals are percentile. The same indices drive statistics within each period; all draws were valid. Fees/trades are not reinserted at resampled block boundaries. Holm3 corrects only the three periods **within that standalone study**, not the nine-case study or the two directions together.

BTC coefficients: alpha is 365 times the daily arithmetic intercept, not CAGR. Full-calendar HAC7 intervals are descriptive. Cash adjustment subtracts the same rate from strategy and BTC; beta is unchanged. These are BTC-only estimates, **not joint 22-coin regressions**, and are not selection gates.

Preserve the procedural distinction

Momentum: its code/settings seal preceded the first all-22 simulation, but existing verification tests had already decoded packaged panels and recalculated saved-nine cash statistics. Strict pre-panel-access compliance is **false**; the chronology exception and corrections remain in its evidence. **Reversal:** its own freeze preceded the first new-study panel decode; strict pre-access checks passed.

All three historical windows had already been examined in the wider project. These additional studies **are not fresh independent OOS confirmation**. A failed rule here does not prove all momentum or reversal strategies fail. No basket, lookback, cadence, direction or fee sensitivity is promoted after seeing these outcomes.

January–September 2026 (273 days): all predeclared sensitivities retained

Primary remains **20 bps / lag1**. Lag1 is idealized observed-close execution, not a guaranteed one-hour wait; lag2 adds one hour. Each independent study completed **21** scenario-period keys, with no missing or failed slots. These tables retain all seven scenarios for this report's period; no sensitivity becomes a new selection.

Standalone momentum: January–September 2026 (273 days)

Cost bps / lag	Net return: raw / vs cash	Raw Sharpe 365 / 252	Excess Sharpe 365 / 252	Hourly max DD
0 / 1	Raw -16.15% Cash -18.01%	-1.26 / -1.05	-1.43 / -1.19	17.66%
5 / 1	Raw -17.37% Cash -19.21%	-1.37 / -1.14	-1.54 / -1.28	18.79%
10 / 1	Raw -18.58% Cash -20.39%	-1.48 / -1.23	-1.65 / -1.37	19.89%
15 / 1	Raw -19.77% Cash -21.55%	-1.59 / -1.32	-1.76 / -1.46	20.98%
20 / 1	Raw -20.94% Cash -22.70%	-1.70 / -1.41	-1.87 / -1.55	22.06%
40 / 1	Raw -25.47% Cash -27.12%	-2.13 / -1.77	-2.30 / -1.91	26.35%
20 / 2	Raw -20.67% Cash -22.43%	-1.66 / -1.38	-1.83 / -1.52	21.79%

Standalone reversal: January–September 2026 (273 days)

Cost bps / lag	Net return: raw / vs cash	Raw Sharpe 365 / 252	Excess Sharpe 365 / 252	Hourly max DD
0 / 1	Raw 16.57% Cash 13.98%	1.25 / 1.04	1.08 / 0.90	9.57%
5 / 1	Raw 14.87% Cash 12.32%	1.14 / 0.95	0.97 / 0.81	9.95%
10 / 1	Raw 13.19% Cash 10.68%	1.03 / 0.86	0.86 / 0.71	10.34%
15 / 1	Raw 11.54% Cash 9.07%	0.92 / 0.76	0.75 / 0.62	10.72%
20 / 1	Raw 9.91% Cash 7.48%	0.81 / 0.67	0.64 / 0.53	11.10%
40 / 1	Raw 3.63% Cash 1.33%	0.36 / 0.30	0.19 / 0.16	14.76%
20 / 2	Raw 9.54% Cash 7.11%	0.78 / 0.64	0.61 / 0.50	11.38%

Full universe, data and supporting evidence

Fixed historical-liquidity slots: BTC, ETH, SHIB, BNB, SOL, XRP, DOT, SAND, legacy LUNA/LUNC, AVAX, DOGE, ADA, FTM/S, TRX, MATIC/POL, MANA, FIL, GALA, LINK, AXS, LTC and VET. Eligible members are reranked together; BTC/ETH are ordinary members, not fixed sleeves. No reconstitution. Inherited inventory/survivor limitations remain.

Source: Binance Data Vision spot-USDT hourly archives; **43,824** normalized endpoints from **1 October 2021 01:00 UTC through 1 October 2026 00:00 UTC**, including warm-up, with seven pinned panels and explicit lifecycle/halt masks. No data were downloaded, gaps filled or windows shortened. Full data manifest. Shorts are theoretical; borrow, funding, margin, impact, capacity, realistic fills and ordinary drift-maintenance are unmodelled. USDT/USD parity and the hypothetical cash proxy are assumptions.

Complete three-period evidence: Momentum results, scenarios, inference, protocol. Reversal results, scenarios, inference, protocol.

1. All nine frozen cases remain in the primary result

The primary convention is **20 bps one-way full-L1 transaction cost at lag 1**, including entry, forced-event and terminal liquidation charges. Lag 1 is an **idealized observed-close fill**, not an actual extra one-hour wait. The window is **1 January–30 September 2026**, ending **1 October at 00:00 UTC**, with 6,552 hourly intervals and 273 complete UTC days.

Returns are simulated nine-month compounded changes. Sharpe, volatility, turnover and CAGR are annualized estimates; the CAGR column is **not a realized full-year return**. Every case begins the evaluation independently from cash and is liquidated once at the final endpoint.

All nine: primary 20 bps results, raw and cash-excess comparison

ID	Gross return	Net return: raw / vs cash	Gross Sharpe (365 / 252)	Net Sharpe (365 / 252): raw / excess	CAGR estimate: raw / vs cash	Original hourly Max DD	Turnover / year
A01	28.80%	Raw 22.24% Cash 19.53%	1.47 / 1.22	Raw 1.19 / 0.99 Cash 1.07 / 0.89	Raw 30.80% Cash 26.93%	14.05%	34.93
A02	24.64%	Raw 14.36% Cash 11.82%	1.26 / 1.05	Raw 0.82 / 0.68 Cash 0.70 / 0.59	Raw 19.65% Cash 16.11%	15.51%	57.49
A03	38.33%	Raw 31.38% Cash 28.46%	2.74 / 2.27	Raw 2.32 / 1.93 Cash 2.13 / 1.77	Raw 44.03% Cash 39.77%	7.28%	34.43
A04	-6.75%	Raw -16.93% Cash -18.77%	0.08 / 0.06	Raw -0.22 / -0.18 Cash -0.28 / -0.23	Raw -21.96% Cash -24.27%	38.89%	77.21
B01	-6.25%	Raw -18.98% Cash -20.78%	-0.15 / -0.12	Raw -0.82 / -0.68 Cash -0.92 / -0.76	Raw -24.53% Cash -26.76%	33.80%	97.60
B03	-41.79%	Raw -45.36% Cash -46.57%	-2.35 / -1.95	Raw -2.65 / -2.20 Cash -2.75 / -2.29	Raw -55.43% Cash -56.75%	45.78%	42.28
B04	-11.64%	Raw -14.29% Cash -16.19%	-0.56 / -0.47	Raw -0.73 / -0.61 Cash -0.86 / -0.71	Raw -18.63% Cash -21.04%	31.80%	20.39
B05	-3.76%	Raw -5.86% Cash -7.95%	-0.09 / -0.07	Raw -0.21 / -0.17 Cash -0.33 / -0.28	Raw -7.75% Cash -10.48%	22.42%	14.71
B06	-0.004%	Raw -4.22% Cash -6.35%	0.19 / 0.16	Raw 0.05 / 0.04 Cash -0.03 / -0.02	Raw -5.60% Cash -8.39%	19.51%	28.75

Raw = original strategy results **net of 20 bps fees**, before cash. **Cash** = wealth-relative return W(strategy)/W(cash) − 1, relative CAGR or daily arithmetic-excess Sharpe (each365 / 252), **not gross versus net fees**. Benchmark: hypothetical 3% nominal USDT cash proxy, c = 0.03/365 daily; USD interpretation assumes parity. No strategy or collateral interest is credited. Hourly DD, turnover and acceptance flags are unchanged. Full methods and intervals.

2. Frozen identities and allocations did not change

Track A contains the four development-only selections; Track B contains five validation-informed family selections. Track B validation Sharpe was therefore a **selection statistic**, not an unbiased final-test estimate. B02 is an empty factor-family slot, not a tenth traded strategy.

Signals use **trailing returns**, not moving-average crossings. Weekly decisions are Monday 00:00 UTC; 72-hour decisions follow the inherited elapsed-time epoch. The 0.25 buffer skips the **entire** rebalance below that proposed full-L1 change. Rank/demean/L1 cross-sectional weights have unit gross exposure and zero net target; trend slot budgets remain 1/8 without rescaling active coins. Legacy symbols denote audited economic slots: **LUNA/LUNC is original Terra, not Terra 2**; FTM/S and MATIC/POL include migration treatment.

Frozen basket membership and settings

ID	Basket	Strategy	Lookback days	Rebalance hours	Buffer	Coins
A01	4,645	Buffered momentum	28	168	0.25	BTC, ETH, SOL, LUNA/LUNC, FTM/S, TRX, LINK, LTC
A02	4,645	Momentum	28	72	0.00	BTC, ETH, SOL, LUNA/LUNC, FTM/S, TRX, LINK, LTC
A03	1,159	Reversal	60	168	0.00	BTC, ETH, SHIB, DOT, ADA, MANA, FIL, VET
A04	4,444	Trend long/short	28	24	0.00	BTC, ETH, SOL, LUNA/LUNC, AVAX, FTM/S, FIL, VET
B01	2,323	Buffered momentum	28	24	0.25	BTC, ETH, BNB, XRP, DOGE, FTM/S, AXS, VET
B03	2,457	Momentum	90	72	0.00	BTC, ETH, BNB, XRP, FTM/S, MATIC/POL, AXS, LTC
B04	6,064	Reversal	90	168	0.00	BTC, ETH, DOT, LUNA/LUNC, AVAX, MANA, FIL, VET
B05	2,330	Trend long/cash	28	168	0.00	BTC, ETH, BNB, XRP, DOGE, TRX, GALA, LINK
B06	2,495	Trend long/short	28	168	0.00	BTC, ETH, BNB, XRP, TRX, MANA, GALA, LINK

Exposure and secondary risk measures

Exposure is measured from held accounting-interval weights. The long/cash case B05 retains inactive signal allocations as cash; unavailable cells are not zero. BTC alpha and beta are descriptive regressions of daily returns, not proof of arbitrage. Alpha, Sharpe and CAGR answer different questions; inherited regression significance is not a new selection gate.

Exposure, volatility and BTC alpha: unchanged raw / cash-excess

ID	Ann. volatility	Gross exposure	Net exposure	Cash allocation	BTC beta	BTC alpha / year: raw / cash
A01	25.22%	100.00%	0.00%	Not applicable	-0.12	Raw 30.54% Cash 27.17%
A02	25.97%	100.00%	0.00%	Not applicable	-0.11	Raw 21.78% Cash 18.43%
A03	16.33%	100.00%	0.00%	Not applicable	0.03	Raw 37.71% Cash 34.79%
A04	52.09%	99.82%	-3.30%	Not applicable	-0.38	Raw -9.75% Cash -13.90%
B01	29.28%	100.00%	0.00%	Not applicable	-0.05	Raw -23.71% Cash -26.85%
B03	28.91%	100.00%	0.00%	Not applicable	-0.11	Raw -76.05% Cash -79.39%
B04	24.11%	100.00%	0.00%	Not applicable	0.03	Raw -17.83% Cash -20.75%
B05	24.43%	49.22%	49.22%	50.78%	0.31	Raw -6.43% Cash -8.50%
B06	39.53%	100.00%	-1.56%	Not applicable	-0.37	Raw 3.52% Cash -0.59%

Like-clock risk: raw daily versus cash-relative daily drawdown

ID	Raw DAILY Max DD	Cash-relative DAILY Max DD
A01	12.72%	12.82%
A02	14.01%	14.36%
A03	6.05%	6.21%
A04	38.65%	39.02%
B01	33.10%	34.33%
B03	45.36%	46.57%
B04	30.28%	31.53%
B05	21.62%	22.89%
B06	18.78%	20.25%

Like-clock interpretation. Raw results remain net of 20 bps transaction costs. Raw and cash-relative daily drawdowns both use daily NAV endpoints including the initial NAV. They are not comparable with the original hourly Max DD as evidence of improved risk. BTC alpha pairs use the same full-calendar HAC7 convention; betas and exposures are unchanged. Confidence intervals, joint basket factors and complete benchmark methods.

3. Positive return and statistical support are separate criteria

Economic: net compounded return > 0. **Statistical:** positive HAC mean-return t-statistic and primary Holm-9 p <= 0.05. **Drawdown:** observed-hour maximum drawdown <= 40%. **Turnover:** annualized full-L1 turnover <= 1.5 times the greater historical development/validation turnover, with caps fixed before this study's access.

These are **evaluation limits, not trading stop-loss switches.** A01 and A02 gained but did not pass the corrected statistical criterion. A04 and all five Track B cases lost. B03 also exceeded its drawdown and turnover limits. No unsuccessful case was removed and no risk cap was revised.

Approved criteria and frozen turnover caps

ID	HAC t	Bootstrap p	Holm-9 p	Economic	Statistical	DD	Turnover	TO cap / year	All criteria
A01	1.56	0.0288	0.2304	Pass	Fail	Pass	Pass	52.72	Fail
A02	0.94	0.1422	0.9953	Pass	Fail	Pass	Pass	83.15	Fail
A03	3.08	0.0024	0.0216	Pass	Pass	Pass	Pass	41.44	Pass
A04	-0.20	0.5774	1.0000	Fail	Fail	Pass	Pass	95.87	Fail
B01	-0.70	0.7046	1.0000	Fail	Fail	Pass	Pass	122.53	Fail
B03	-3.05	0.9996	1.0000	Fail	Fail	Fail	Fail	41.63	Fail
B04	-0.62	0.7231	1.0000	Fail	Fail	Pass	Pass	33.52	Fail
B05	-0.19	0.5767	1.0000	Fail	Fail	Pass	Pass	18.72	Fail
B06	0.07	0.4604	1.0000	Fail	Fail	Pass	Pass	36.89	Fail

Uncertainty uses the shared calendar, not independent daily observations

The primary hypothesis is positive **expected arithmetic daily net return**, not positive BTC alpha or geometric growth. HAC uses a Bartlett kernel with 27 lags and the frozen n/(n-1) variance adjustment. The primary test uses **10,000 centered circular 28-day block-bootstrap-t samples**, seed 20261015, synchronized across all nine strategies; Holm retains all nine hypotheses.

Daily-mean intervals are 95% studentized intervals; Sharpe and annualized complete-day geometric-growth intervals are 95% percentile intervals. These are **approximate marginal intervals**, not jointly 95% across nine cases. The separately shown familywise mean lower bound is Bonferroni, **not a Holm-inverted interval or another success gate**. Only about 9.75 block lengths fit this nine-month sample, so dependence/stationarity assumptions and limited power matter. Holm-9 does not correct the much larger earlier research search.

Approximate intervals; no OOS reselection

ID	Mean daily	Mean 95% CI	Sharpe 95% CI (365 / 252)	Growth 95% CI	Familywise mean LB
A01	0.082%	[-0.003, 0.186]%	365 [-0.28, 2.58] 252 [-0.23, 2.14]	[-8.93, 93.45]%	-0.027%
A02	0.058%	[-0.045, 0.179]%	365 [-0.87, 2.17] 252 [-0.72, 1.80]	[-19.56, 78.74]%	-0.073%
A03	0.104%	[0.039, 0.171]%	365 [0.85, 3.88] 252 [0.71, 3.22]	[12.83, 86.64]%	0.013%
A04	-0.031%	[-0.343, 0.347]%	365 [-2.72, 1.65] 252 [-2.26, 1.37]	[-71.70, 119.50]%	-0.452%
B01	-0.065%	[-0.218, 0.182]%	365 [-3.50, 1.25] 252 [-2.90, 1.04]	[-59.14, 46.84]%	-0.254%
B03	-0.210%	[-0.396, -0.087]%	365 [-5.09, -0.98] 252 [-4.23, -0.81]	[-76.94, -26.69]%	-0.453%
B04	-0.049%	[-0.255, 0.110]%	365 [-2.91, 1.88] 252 [-2.42, 1.56]	[-57.41, 44.44]%	-0.337%
B05	-0.014%	[-0.187, 0.145]%	365 [-2.51, 1.92] 252 [-2.09, 1.59]	[-46.83, 55.15]%	-0.241%
B06	0.005%	[-0.128, 0.168]%	365 [-1.53, 1.25] 252 [-1.27, 1.04]	[-42.89, 60.11]%	-0.167%

4. Drawdown dates describe the observed hourly path

Dates and durations refer to the **maximum-depth drawdown episode**. Recovery is the first observed endpoint regaining its previous peak; an unrecovered episode is right-censored at 1 October 00:00 UTC. The maximum-depth episode need not be the longest underwater episode, which is retained separately below. Intrahour drawdowns are unobserved.

Daily expected shortfall (ES95) is the mean loss in the worst 5% of complete daily observations, using the established engine convention. These risk diagnostics do not change the frozen decision rule.

Maximum-depth episodes; all dates UTC

ID	Peak UTC	Trough UTC	Recovery UTC	To trough days	DD episode days
A01	2026-05-05 18:00	2026-05-16 09:00	2026-09-06 20:00	10.63	124.08
A02	2026-02-28 20:00	2026-04-20 16:00	2026-05-03 00:00	50.83	63.17
A03	2026-01-18 01:00	2026-01-31 18:00	2026-02-25 18:00	13.71	38.71
A04	2026-02-06 01:00	2026-04-20 00:00	Unrecovered	72.96	236.96
B01	2026-01-23 15:00	2026-09-18 02:00	Unrecovered	237.46	250.38
B03	2026-01-01 04:00	2026-09-30 07:00	Unrecovered	272.13	272.83
B04	2026-01-23 05:00	2026-08-31 16:00	Unrecovered	220.46	250.79
B05	2026-01-14 17:00	2026-06-06 05:00	Unrecovered	142.50	259.29
B06	2026-02-06 00:00	2026-09-16 18:00	Unrecovered	222.75	237.00

Additional downside and duration measures

ID	Longest underwater days	Worst day	Daily ES95 loss
A01	124.08	-4.66%	2.65%
A02	139.58	-5.64%	2.65%
A03	38.71	-2.34%	1.57%
A04	236.96	-11.26%	6.03%
B01	250.38	-7.42%	3.27%
B03	272.83	-7.52%	3.91%
B04	250.79	-5.37%	3.00%
B05	259.29	-5.28%	3.09%
B06	237.00	-10.62%	4.80%

Full indicator companion

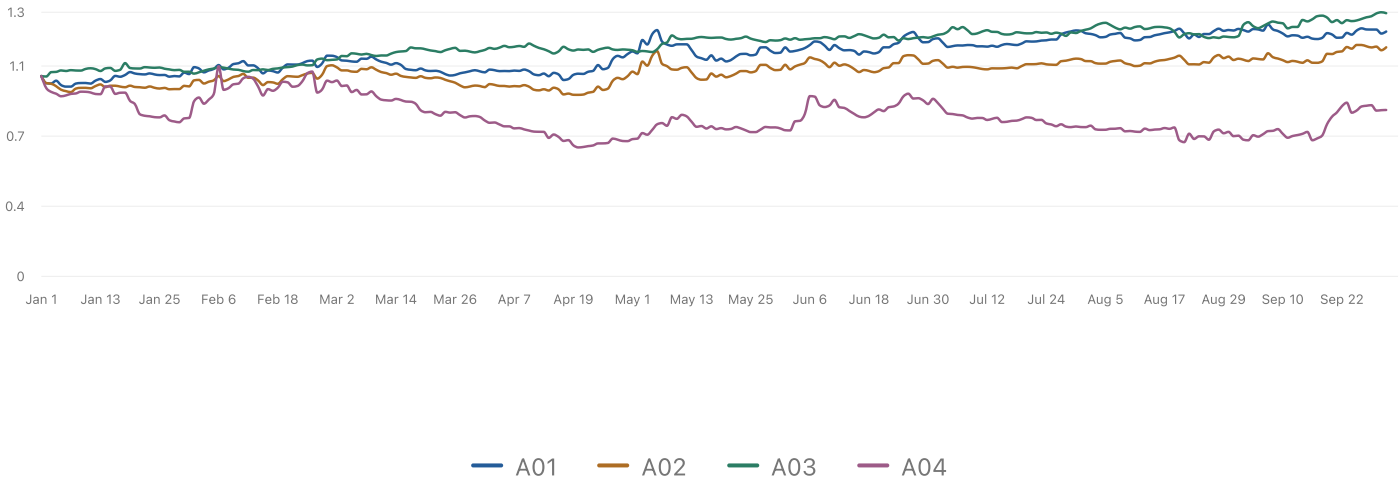
Download all nine primary indicators, including Sortino, Calmar, ulcer index, time underwater, daily win rate, tail measures and full BTC attribution. No trade-level profit factor is fabricated without a trade-grouping definition.

5. Neither track was consistently successful across periods

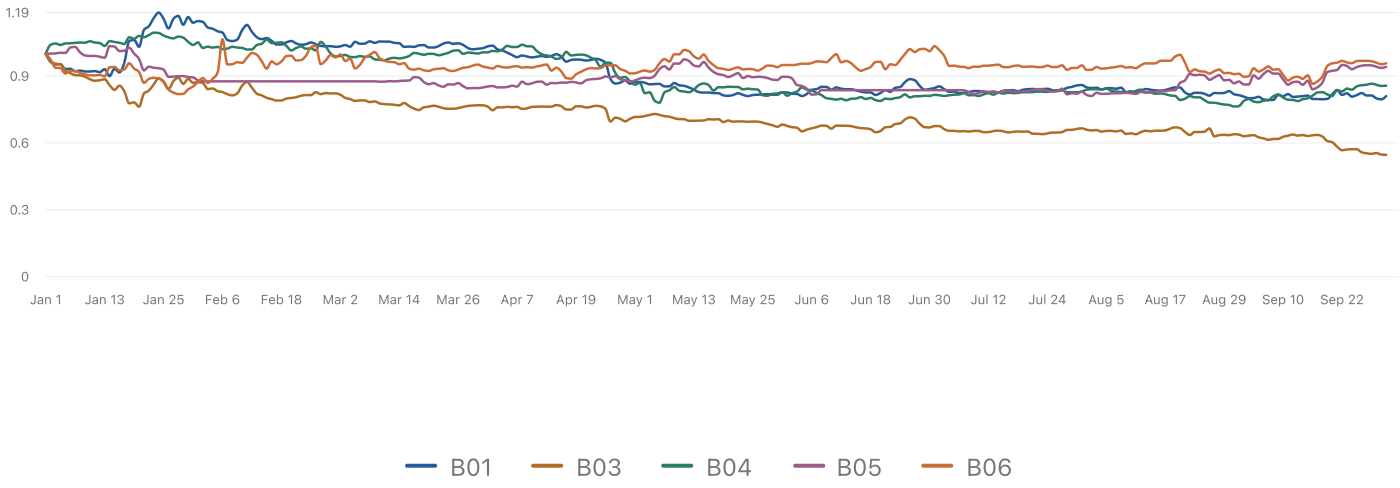
The plots show each saved path's daily closing net NAV, starting at 1.00. They are **not combined portfolios**; no portfolio weights are inferred. All daily observations and the terminal liquidation endpoint are retained. Earlier development/validation windows each span two years, whereas this evaluation covers nine months, so raw returns are not equal-horizon rates.

A03 passed this test but lost in validation; Track B's favorable validation did not carry through to 2026. These observations do not constitute a formal claim that Track A is generally superior.

Track A: daily closing net NAV, start = 1



Track B: daily closing net NAV, start = 1



Same-pair comparisons across all periods

Download the 27 same-pair period rows. Development, validation and partial-year evaluation remain distinct; no period has been moved to improve a result.

6. Fee sensitivities do not replace the primary test

All **63 frozen case-scenario outcomes** remain available: 0/5/10/15/20/40 bps at lag 1 and 20 bps at lag 2. Lag 2 adds one hour. Full-L1 charges include entry and terminal exit; ordinary drift-maintenance trades, borrow, funding and realistic fills are omitted.

A03 remains positive at **40 bps (+24.77%)** and with the additional-hour delay (**+33.01%**), versus +31.38% primary. These are diagnostics, not fee-optimized rescue tests. B06 has slightly positive arithmetic daily Sharpe (**0.05 at 365-day / 0.04 at 252-day**) but negative compounded return (**-4.22%**); arithmetic and geometric measures can differ without an accounting contradiction.

Net period returns under every frozen fee/delay scenario

ID	0 bps	5 bps	10 bps	15 bps	20 bps	40 bps	20 bps +1h
A01	28.80%	27.13%	25.48%	23.85%	22.24%	16.00%	23.07%
A02	24.64%	21.98%	19.39%	16.85%	14.36%	4.91%	13.06%
A03	38.33%	36.56%	34.81%	33.08%	31.38%	24.77%	33.01%
A04	-6.75%	-9.40%	-11.98%	-14.49%	-16.93%	-26.01%	-16.75%
B01	-6.25%	-9.61%	-12.85%	-15.97%	-18.98%	-30.00%	-24.40%
B03	-41.79%	-42.70%	-43.60%	-44.49%	-45.36%	-48.71%	-47.03%
B04	-11.64%	-12.31%	-12.97%	-13.64%	-14.29%	-16.87%	-13.83%
B05	-3.76%	-4.29%	-4.82%	-5.34%	-5.86%	-7.91%	-4.34%
B06	-0.004%	-1.08%	-2.14%	-3.18%	-4.22%	-8.27%	-1.12%

7. Post-evaluation checks locate sensitivity, not a replacement winner

A03's three nearest pre-enumerated parameter controls all gained in 2026, but their validation results ranged from **-43.76% to +3.75%**. All **11 Track B parameter neighbors** lost in 2026. Thus the observed failures were not confined to one precise setting. The counts below retain every frozen anchor, without ranking the controls.

Basket sensitivity remained substantial: applying B04's same rule to the **predefined liquidity-tier reference basket** returned **+11.17%**, versus **-14.29%** on its selected basket. Several members change together; this is not an isolated coin effect and the reference is not promoted.

A03's short contribution reversed sign: **-26.95 percentage points in validation versus +10.86 pp in 2026**. Its long side contributed +20.52 pp in 2026. Contributions are linked hourly net P&L in points of initial capital and reconcile to the original compounded path, not independently compounded coin or sleeve returns.

A03 attribution: linked contribution, not independent sleeve return

Period	Net return	Long contribution pp	Short contribution pp	Largest abs. coin share	Best 5 weeks share	Positive months
2022-2023	64.38%	32.69	31.69	27.07%	35.64%	14/24
2024-2025	-22.31%	4.64	-26.95	29.00%	52.53%	9/24
Jan-Sep 2026	31.38%	20.52	10.86	19.55%	51.21%	7/9

Positive / all pre-enumerated nearest-parameter controls

ID	2022-2023	2024-2025	Jan-Sep 2026
A01	2/2	0/2	1/2
A02	4/4	0/4	3/4
A03	3/3	2/3	3/3
A04	2/2	1/2	2/2
B01	1/2	2/2	0/2
B03	1/3	3/3	0/3
B04	0/2	2/2	0/2
B05	0/2	2/2	0/2
B06	0/2	1/2	0/2

Same-rule reference-basket contrasts

Download all posthoc control contrasts. The reference-basket and nearest-parameter comparisons are explanatory, not a new selection exercise.

Why there is no B02

The factor-residual family had **1,355 development-Q1 baskets**; **740** had positive development returns, but **zero** also gained in validation with defined Sharpe. Across all 6,615 factor baskets, none gained in validation even before costs. **6,015 did trade** during validation, while 600 were cash: absence of a shortlist candidate is not the same as universal monthly model ineligibility. No substitute factor candidate was manufactured.

Factor-family qualification funnel

Stage	Pairs
All factor baskets	6,615
Development Q1	1,355
Evaluated in both periods	1,355
Positive development	740
Positive both periods	0
Eligible with defined Sharpes	0

8. The full exploratory cohort did not demonstrate robust transfer

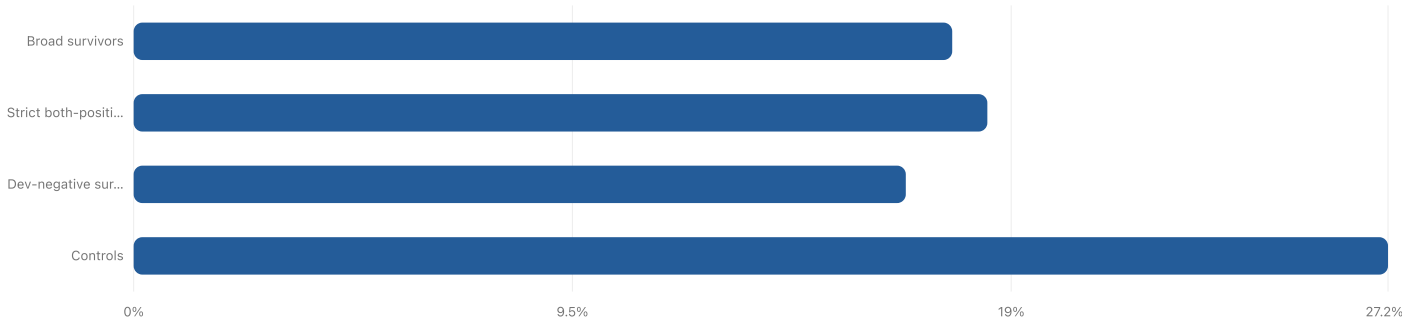
Q1 means approximately the best 20% of defined development Sharpe ranks within each basket across 169 configurations, with average ties retained—not calendar quarter one. The full Q1 census has 224,867 basket-configuration pairs. Its 69,108 validation-positive survivors split into 39,332 positive in both periods and 29,776 development-negative survivors; the remaining 155,759 are validation-nonpositive controls.

The middle two rows below **partition**, rather than add to, broad survivors. These are constituent distributions, **not investable portfolio returns or independent trials.** The subgroup definition and screening tests were frozen for an exploratory review of already-examined 2026 observations, not an independent new test. All 639,515 requested scenario rows were evaluated; no unfavorable or missing result was silently dropped.

Exploratory 2026 constituent distributions, 20 bps

Cohort	Pairs	Profitable	Median net return	Median Sharpe (365 / 252)	Median Max DD	Median TO / year
Broad survivors	69,108	17.72%	-12.08%	-0.42 / -0.35	28.84%	47.80
Strict both-positive	39,332	18.48%	-10.95%	-0.32 / -0.27	29.09%	39.61
Dev-negative survivors	29,776	16.72%	-13.83%	-0.59 / -0.49	28.43%	50.81
Controls	155,759	27.16%	-10.82%	-0.43 / -0.36	25.85%	68.86

Profitable fraction: groups are overlapping, not additive



All cohort fee and delay sensitivities

The source view retains every predeclared cohort fee/delay scenario. Broad and strict median zero-fee returns were still negative. All 69,108 survivor rows and all 39,332 strict-subgroup rows remain in experiment-ID/basket-ID order, not OOS performance order.

Dependence-aware screening comparisons remain inconclusive

The statistic is survivors-minus-controls in the fraction with **strictly positive full-period net return**. Four predeclared contrasts use **4,000 synchronized circular 28-day blocks**, PCG64 seed 20261015. Every constituent shares each calendar draw. Intervals are 95% marginal percentile intervals; centered two-sided bootstrap p-values use the +1 convention and Holm correction across **these four contrasts only**.

The configuration-standardized contrast uses **62 shared configurations and 207,804 Q1 pairs**, weighted by fixed past all-Q1 case mass. It is -10.71 pp, with **[-16.77, +1.21] pp** interval and Holm p = **0.237**. All four intervals include zero. This does not prove screening harmful, equivalent to no screening, or useless; the grid is highly dependent and the correction is not research-wide search adjustment.

Four exploratory screening comparisons; sign = survivors minus controls

Contrast	Lift pp	95% CI pp	Centered p	Holm-4 p	Reject 5%
Overall survivors - controls	-9.44	[-18.21, 9.24]	0.3224	0.6448	Fail
Within development-positive	-9.37	[-20.73, 15.25]	0.4039	0.6448	Fail
Within development-negative	-9.92	[-18.52, 7.33]	0.2002	0.6006	Fail
Configuration-standardized	-10.71	[-16.77, 1.21]	0.0592	0.2369	Fail

9. Controls' advantage began before modeled transaction costs

Controls had **50.69%** gross-profitable paths versus **33.06%** among survivors: a **17.63 pp** advantage. Costs erased profit in **23.53%** of all control paths and **15.33%** of survivor paths, narrowing the gap by **8.20 pp** to **9.44 pp** net. Controls traded **more**, not less: median annualized full-L1 turnover was **68.86x versus 47.80x**.

Even at zero transaction cost, broad and strict survivors had negative median compound returns (**-5.96% and -5.22%**). Signal/path transfer, not only fees, was weak. Separate medians need not describe the same constituent path.

Controls traded more but had stronger pre-fee breadth

Cohort	Gross profitable	Net profitable	Gross profit erased pp	Median gross return	Median net return	Median TO / year
Survivors	33.06%	17.72%	15.33	-5.96%	-12.08%	47.80
Controls	50.69%	27.16%	23.53	0.22%	-10.82%	68.86

Family mix matters, but lower turnover was not sufficient

Reversal represented **27.19% of controls versus 4.81% of survivors**; long/cash trend was **3.96% versus 27.04%**, partly offsetting that imbalance. Family composition contributed **2.33 pp** to the raw net gap, with **6.94 pp within shared families** and **0.17 pp** from controls-only residual cases. The advantage was not universal: reversal survivor breadth was **49.68% versus 48.03%** for reversal controls.

Survivors were more often weekly (**56.23% versus 42.87%**) and in 14/28-day lookbacks (**70.50% versus 41.23%**). Among 14-day long/cash survivors, profitable breadth was **52.62% daily versus 18.21% weekly**; for long/short it was **52.97% versus 6.23%**. These contain different basket mixtures, not causal parameter experiments. No setting is selected from these observations.

Every family: no favorable family is promoted

Family	Cohort	Pairs	Share of cohort	Profitable	Median net return	Median Sharpe (365 / 252)
Buffered momentum	Survivors	6,879	9.95%	4.32%	-22.07%	-1.39 / -1.16
Buffered momentum	Controls	13,189	8.47%	10.97%	-18.83%	-1.08 / -0.90
Momentum	Survivors	29,269	42.35%	10.00%	-19.38%	-1.11 / -0.92
Momentum	Controls	78,143	50.17%	17.76%	-16.65%	-0.93 / -0.77
Reversal	Survivors	3,323	4.81%	49.68%	-0.06%	0.10 / 0.08
Reversal	Controls	42,350	27.19%	48.03%	-0.92%	0.06 / 0.05
Trend long/cash	Survivors	18,685	27.04%	28.98%	-4.58%	-0.06 / -0.05
Trend long/cash	Controls	6,171	3.96%	41.58%	-1.81%	0.08 / 0.07
Trend long/short	Survivors	10,952	15.85%	17.90%	-11.57%	-0.13 / -0.11
Trend long/short	Controls	14,551	9.34%	26.23%	-8.40%	-0.02 / -0.02
BTC-factor residual	Controls	1,355	0.87%	18.97%	-0.41%	-0.67 / -0.56

Settings companion

Download all settings and subgroups, including the full eight trend settings and every represented configuration. An unavailable factor-survivor cell means no qualifying member, not zero performance.

10. Basket composition explains part, not all, of the difference

The legacy **LUNA/LUNC** slot appeared in **33.47% of controls versus 17.53% of survivors**; XRP appeared in **23.64% versus 39.68%**, and legacy FTM/S in **24.41% versus 37.97%**. BTC and ETH were fixed in every basket, so their inclusion effects cannot be estimated here. Coin associations involve within-tier replacement baskets, not isolated coin P&L or recommendations.

Each row below is an **alternative** decomposition of the same 9.44 pp raw control advantage. Within a row, composition + within-stratum + unsupported terms reconcile. **Do not add rows together.** Exact-configuration standardization still compares different baskets; exact-basket standardization still compares different strategies. Unsupported groups remain explicit, without fabricated missing-cohort rates.

All 22 economic slots were active in 2026, which rules out **retired-slot cash** as the explanation. It does not rule out ordinary signal-driven cash in long/cash trend.

Alternative standardizations; all gaps are controls minus survivors

Conditioning	Composition pp	Within pp	Unsupported pp	Raw gap pp	Standardized gap pp	Shared strata	Shared pairs
Family	2.33	6.94	0.16	9.44	6.34	5	223,512
Exact configuration	-4.24	11.02	2.65	9.44	10.71	62	207,804
Exact basket	3.81	5.59	0.04	9.44	4.92	6,611	224,731

A joint additive sensitivity leaves a descriptive 5.97 pp gap

An additive configuration-plus-basket fixed-effect projection on **207,674 pairs**, 62 shared configurations and 6,611 shared baskets leaves a **5.97 pp** controls-minus-survivors coefficient, versus **9.79 pp** raw within that retained subset. Independent sparse solvers agree.

This assumes common additive basket effects despite possible strategy-basket interactions. It has **no causal interpretation, standard error or p-value** and is not an extra term to add to the decompositions. Exact conditioning on both configuration and basket would identify the original pair itself; a pair cannot be both survivor and control.

Joint additive sensitivity; not causal or an independent hypothesis test

Pairs	Configurations	Baskets	Raw retained gap pp	Adjusted gap pp
207,674	62	6,611	9.79	5.97

Basket composition companion

Download historical-rank coin inclusion frequencies. Historical liquidity rank is not a 2026 performance rank; fixed BTC/ETH inclusion has no identifiable contrast here.

11. Market conditions changed, but regime causality is unproven

Validation's BTC/ETH endpoint returns were **+107.29% / +30.23%**, versus **-4.59% / -9.61%** in January-September 2026. Median trailing BTC volatility fell from **45.01% to 40.66%**. Endpoint losses do not imply continuously falling prices.

Regimes use information available at each **00:00 UTC decision**, applied to the **following day**. High/low median thresholds are fitted using 2022-2023 only; BTC direction is the sign of its trailing 30-day return. Unknown regimes are retained; no strategy switches on/off and no Sharpe is computed from discontinuous concatenated regime observations. The fixed-22 contextual 28-day rank IC was only **-0.019** in 2026 and is not the engine's eight-coin signal.

Some raw regime comparisons changed sign after configuration standardization: settings mix confounded apparent state effects. The table retains raw and same-setting daily-mean gaps rather than claiming a unique regime cause.

Period context: unequal horizons, not timing signals

Period	Days	BTC hold return	ETH hold return	Median BTC trailing vol	BTC up-state days	High vol days	Context 28d rank IC
2022-2023	730	-8.51%	-37.93%	49.04%	366	365	-0.00
2024-2025	731	107.29%	30.23%	45.01%	401	259	-0.01
Jan-Sep 2026	273	-4.59%	-9.61%	40.66%	159	56	-0.02

Decision-time states versus following-day returns; gaps in bps/day

Regime	State	Days	Raw C-S gap bps/day	Same-setting gap bps/day
BTC 30-day direction	High / up	159	3.36	3.27
BTC 30-day direction	Low / down	114	-2.96	-0.60
BTC 30-day direction	Unknown	0	Unavailable	Unavailable
BTC trailing volatility	High / up	56	1.11	-0.75
BTC trailing volatility	Low / down	217	0.62	2.27
BTC trailing volatility	Unknown	0	Unavailable	Unavailable
Cross-coin dispersion	High / up	108	-1.81	1.33
Cross-coin dispersion	Low / down	165	2.38	1.86
Cross-coin dispersion	Unknown	0	Unavailable	Unavailable
Relative BTC quote volume	High / up	135	-1.68	0.70
Relative BTC quote volume	Low / down	138	3.07	2.58
Relative BTC quote volume	Unknown	0	Unavailable	Unavailable

Regime interactions do not establish reliable outperformance

The interaction is the **controls-minus-survivors mean daily gap in high/up minus the gap in low/down**, in basis points. Three 4,000-draw joint-calendar bootstrap intervals include zero. The volatility CI and estimated p-value are **unavailable** because **29 draws lacked a required state**; they were not dropped or filled.

Holm keeps a four-test family using **p = 1 only as a conservative correction placeholder** for the unavailable volatility hypothesis. The volatility row remains unavailable, not a valid p = 1 finding or proof of no effect. All regime comparisons are posthoc, conditional associations—not causal explanations or strategy-selection evidence.

Regime interactions: missing evidence stays unavailable

Regime	Interaction bps	95% CI bps	Centered p	Holm-4 p	Invalid draws	Status
BTC 30-day direction	6.32	[-0.93, 14.21]	0.1082	0.4329	0	Conditional posthoc
BTC trailing volatility	0.49	Unavailable	Unavailable	Unavailable	29	Unavailable; no state in 29 draws
Cross-coin dispersion	-4.18	[-13.47, 4.05]	0.3559	1.0000	0	Conditional posthoc
Relative BTC quote volume	-4.75	[-16.02, 5.63]	0.3917	1.0000	0	Conditional posthoc

12. Historical statistical support is not execution feasibility

The stress adds **10 bps one-way slippage plus 20% annual simple carry on negative held notional**, on top of 20 bps base charges. A03 remains positive at **+18.80%**. These are **hypothetical assumptions**, not observed funding, borrow, spread or market impact; actual short availability is unverified.

At constant **100,000-USDT scenario capital**, A03's absolute target-change notional divided by preceding completed-hour quote volume was **89.61% at the 95th percentile and 467.29% maximum**. The maximum was a 6,250-USDT MANA target change versus 1,337.49 USDT prior-hour volume. Even at 25k capital the maximum proxy was 116.82%.

This is **not actual participation in a future fill window, order-book impact, certified capacity, an AUM recommendation or proof of impossibility**. It warns against assuming an observed-close fill is harmless. Borrow inventory, funding, historical spread/depth and fill reconstruction remain unresolved; drift-maintenance trades are omitted.

Hypothetical stress and preceding-hour volume proxy; 100k-USDT scenario

ID	Base net return	Stressed net return	P95 volume proxy	Max volume proxy	Unknown volume events
A01	22.24%	10.50%	32.49%	130.40%	0
A02	14.36%	1.64%	39.16%	250.52%	0
A03	31.38%	18.80%	89.61%	467.29%	0
A04	-16.93%	-27.42%	131.67%	765.94%	0
B01	-18.98%	-30.12%	59.16%	267.62%	0
B03	-45.36%	-50.88%	37.13%	130.56%	0
B04	-14.29%	-21.68%	96.04%	213.94%	0
B05	-5.86%	-6.89%	24.13%	50.23%	0
B06	-4.22%	-13.12%	128.72%	292.14%	0

13. Final conclusions

The project produced reproducible conditional findings, not a deployable arbitrage strategy. A03 passed the approved historical nine-case test, but lost in validation and faces material execution uncertainty. Neither a positive case nor profitable controls are promoted. Track B's validation-informed choices failed this partial-year window; the broad validation screen did not demonstrate reliable economic transfer.

The control advantage was already present before transaction costs. Family, exact-setting and basket composition explain portions of the association, but no single parameter, coin or market regime is a proven causal explanation. Wide shared-calendar uncertainty is consistent with limited market history despite a large configuration grid.

The chronology matters: development was **2022-2023**, validation **2024-2025**, and the partial-year evaluation **January-September 2026**. The approved protocol and window amendment fixed this evaluation's rules before its price access. However, **2026 prices had been examined before later diagnostics; those diagnostics are not independent new confirmation or live performance**. Nine-case Holm, four-screening-contrast Holm and four-regime Holm apply only to their stated conditional families, not the entire adaptive project.

No live orders, monitoring or deployment are authorized by these findings. An independent future test would require unseen observations and a separately approved, execution-aware protocol; no such result is claimed here.

Evidence and companion results

This report covers the frozen-nine OOS evaluation, parameter and basket controls, exploratory screening cohort and explanatory gap diagnostics. Reports 1 and 2 cover development/robustness and validation/frozen selection.

Data source and complete study dataset. Binance public Data Vision (**data.binance.vision**) spot-USDT **1-hour klines** supply prices and quote volumes. The normalized UTC grid has **43,824 hourly close endpoints**, from **1 October 2021 01:00 to 1 October 2026 00:00**, including warm-up. The packaged **22-slot screening universe** includes SAND; the nine frozen strategies use the **21-slot union of the full universe except SAND**. Lifecycle and missingness masks remain explicit. Data documentation | Full coin inventory | File hashes and coverage.

The performance tables use audited, saved metric and accounting outputs. Per-coin attribution and joint-factor diagnostics use the same strategy paths and existing factor prices; they do not regenerate signals, targets, trades or selections. The nine strategies have independent accounting/calendar/metric and inference audits. The large cohort has exhaustive coverage/algebra audits plus bounded public-engine parity, not independent signal reimplementation for every pair.

Companion files retain **all nine primary metrics and all 63 scenarios**, same-pair period comparisons, every 69,108 survivor and 39,332 strict case in fixed experiment/basket order, full settings/basket/regime diagnostics, attribution and execution scenarios. These are supporting results, not another report or a ranked shortlist. Self-contained HTML supports offline reading; source actions retain definitions and provenance.

Nine-case inference · All 63 scenarios · Attribution · Hypothetical carry/slippage · Volume proxy · Regime interactions · Full regime results

Reading order: 1. Development and robustness; 2. Validation and frozen selection; 3. This report. Companion CSVs are supporting results, not additional reports.

Per-coin attribution: what each coin contributed, and what joint basket exposure explains

1 January-30 September 2026, all nine unchanged frozen strategies. These explanatory diagnostics separate accounting contribution from statistical sensitivity. Strategy returns, Sharpe, turnover, drawdowns, baskets and decisions remain fixed. These factor diagnostics explain the same evaluation paths; they do not select a new winner or change the nine-case acceptance decisions.

The overview shows **annual alpha in percent per year**, estimate [95% CI]. Alpha is 365 times the daily arithmetic intercept, **not CAGR**. The BTC-only comparison uses its full-calendar HAC7 estimate; matched BTC and the joint model use the identical complete-case dates with calendar HAC27.

January-September 2026: all nine frozen strategies

ID	Joint alpha / yr [95% CI]	BTC alpha / yr [95% CI]	Matched BTC alpha / yr [95% CI]	Fit / calendar days	R^2	Max VIF
A01	9.42 [-28.82, 47.66]	30.54 [-22.56, 83.64]	30.54 [-7.71, 68.79]	273/273	0.424	8.23
A02	3.95 [-42.10, 50.01]	21.78 [-34.44, 77.99]	21.78 [-22.27, 65.83]	273/273	0.370	8.23
A03	40.36 [22.10, 58.62]	37.71 [5.97, 69.44]	37.71 [13.88, 61.53]	273/273	0.331	6.61
A04	-13.46 [-107.87, 80.95]	-9.75 [-122.21, 102.71]	-9.75 [-112.35, 92.85]	273/273	0.193	7.23
B01	-39.74 [-75.51, -3.97]	-23.71 [-95.86, 48.43]	-23.71 [-88.86, 41.43]	273/273	0.238	8.95
B03	-81.53 [-122.99, -40.07]	-76.05 [-136.72, -15.38]	-76.05 [-124.58, -27.52]	273/273	0.290	8.99
B04	-0.05 [-23.01, 22.92]	-17.83 [-72.37, 36.72]	-17.83 [-74.36, 38.70]	273/273	0.735	6.62
B05	-11.34 [-40.36, 17.68]	-6.43 [-46.56, 33.70]	-6.43 [-37.24, 24.37]	273/273	0.496	8.87
B06	-12.77 [-66.45, 40.92]	3.52 [-70.53, 77.57]	3.52 [-51.21, 58.25]	273/273	0.220	8.88

How to read this. The joint regression includes an intercept and all eight basket coins simultaneously. A conditional beta is sensitivity holding the other seven factors fixed; it is not a portfolio weight, a causal contribution or proof of market neutrality. R-squared is an in-sample descriptive fit. Comparing joint with matched BTC separates added factors from sample exclusions; the full-calendar BTC-only comparison also uses a different covariance convention.

Uncertainty and missingness. Joint and matched BTC intervals are approximate marginal 95% normal-Wald intervals using Bartlett calendar HAC27 and n/(n-k) correction; HAC7 sensitivity is exported. Missing factor days are excluded from regressions only; gaps are not filled or compressed. The complete strategy and attribution calendars remain intact. Standardized condition >30 or any VIF >10 is a warning, not a reason to drop a coin. Some fits are sensitive to extreme days; high leverage/Cook's distance can coexist with modest collinearity. No event is removed and no model reselected.

Per-coin accounting. Gross contributions and positive allocated fees are both linked with prior **net** strategy NAV; net = gross - fees. Units are whole-period percentage points (pp) of initial capital, not standalone coin returns. The next pages retain all eight slots and their marginal net-contribution intervals.

A01 | basket 4645 | January-September 2026

Buffered momentum; 28.0d lookback, 168h rebalance. **Net period return: 22.24%.** The eight net contributions below sum to that unchanged return.

Joint fit **273/273 days**; R-squared **0.424**; standardized condition **8.31**; max VIF **8.23**. Neither declared collinearity warning threshold was crossed. Max leverage **0.334**, max Cook's distance **0.471**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.123 [-0.192, -0.053]**; matched-calendar HAC27 **-0.123 [-0.215, -0.031]**. All brackets are marginal 95% intervals.

A01: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	2.83	0.62	2.21	[-8.55, 10.77]	-0.142 [-0.274, -0.011]
ETH	11.45	0.67	10.79	[-4.54, 26.23]	0.017 [-0.172, 0.207]
SOL	-4.90	0.75	-5.65	[-19.76, 6.24]	-0.007 [-0.162, 0.147]
LUNA/LUNC	14.38	0.78	13.60	[-17.96, 65.87]	0.147 [0.079, 0.216]
FTM/S	4.01	0.71	3.30	[-24.17, 38.51]	-0.124 [-0.197, -0.051]
TRX	-1.22	0.82	-2.04	[-10.23, 4.88]	0.115 [0.015, 0.216]
LINK	0.22	0.66	-0.44	[-14.11, 16.90]	0.065 [-0.056, 0.186]
LTC	1.26	0.79	0.48	[-8.94, 12.53]	-0.086 [-0.252, 0.080]

A02 | basket 4645 | January-September 2026

Momentum; 28.0d lookback, 72h rebalance. **Net period return: 14.36%.** The eight net contributions below sum to that unchanged return.

Joint fit **273/273 days**; R-squared **0.370**; standardized condition **8.31**; max VIF **8.23**. Neither declared collinearity warning threshold was crossed. Max leverage **0.334**, max Cook's distance **0.336**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.115 [-0.198, -0.032]**; matched-calendar HAC27 **-0.115 [-0.228, -0.002]**. All brackets are marginal 95% intervals.

A02: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	1.06	0.84	0.21	[-9.68, 8.40]	-0.120 [-0.284, 0.044]
ETH	5.43	1.04	4.39	[-6.93, 16.77]	-0.002 [-0.201, 0.198]
SOL	-5.05	1.02	-6.07	[-18.90, 4.44]	-0.021 [-0.190, 0.148]
LUNA/LUNC	16.87	0.93	15.94	[-10.49, 61.01]	0.160 [0.084, 0.236]
FTM/S	5.80	1.22	4.58	[-21.25, 34.75]	-0.090 [-0.184, 0.004]
TRX	-4.36	1.45	-5.80	[-13.31, -0.11]	0.031 [-0.037, 0.098]
LINK	-1.16	1.10	-2.26	[-18.63, 19.05]	0.049 [-0.069, 0.167]
LTC	4.51	1.15	3.36	[-6.64, 15.94]	-0.078 [-0.237, 0.082]

A03 | basket 1159 | January-September 2026

Reversal; 60.0d lookback, 168h rebalance. **Net period return: 31.38%.** The eight net contributions below sum to that unchanged return.

Joint fit **273/273 days**; R-squared **0.331**; standardized condition **8.28**; max VIF **6.61**. Neither declared collinearity warning threshold was crossed. Max leverage **0.358**, max Cook's distance **0.217**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **0.028 [-0.030, 0.086]**; matched-calendar HAC27 **0.028 [-0.039, 0.094]**. All brackets are marginal 95% intervals.

A03: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	3.50	0.51	2.98	[-11.53, 20.48]	-0.077 [-0.186, 0.032]
ETH	-9.03	0.69	-9.73	[-26.83, 12.31]	-0.090 [-0.185, 0.004]
SHIB	8.88	0.88	8.00	[-4.24, 22.46]	-0.038 [-0.087, 0.011]
DOT	4.22	0.69	3.54	[-15.58, 21.85]	0.021 [-0.011, 0.053]
ADA	14.16	0.92	13.24	[1.99, 26.62]	0.083 [0.027, 0.138]
MANA	16.53	0.71	15.82	[0.93, 37.04]	0.063 [-0.013, 0.139]
FIL	13.36	0.80	12.56	[-10.81, 34.43]	0.077 [-0.002, 0.157]
VET	-14.30	0.75	-15.05	[-36.87, 4.69]	-0.025 [-0.115, 0.065]

A04 | basket 4444 | January-September 2026

Trend long/short; 28.0d lookback, 24h rebalance. **Net period return: -16.93%.** The eight net contributions below sum to that unchanged return.

Joint fit **273/273 days**; R-squared **0.193**; standardized condition **7.95**; max VIF **7.23**. Neither declared collinearity warning threshold was crossed. Max leverage **0.411**, max Cook's distance **0.391**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.383 [-0.782, 0.017]**; matched-calendar HAC27 **-0.383 [-0.875, 0.109]**. All brackets are marginal 95% intervals.

A04: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	1.52	1.05	0.47	[-4.33, 10.62]	-0.048 [-0.710, 0.615]
ETH	2.49	1.07	1.42	[-4.94, 15.47]	-0.431 [-0.843, -0.018]
SOL	-3.36	1.17	-4.53	[-11.18, 7.31]	-0.027 [-0.257, 0.204]
LUNA/LUNC	-3.36	1.16	-4.52	[-24.09, 18.42]	0.094 [0.014, 0.175]
AVAX	-0.99	1.61	-2.61	[-13.48, 17.76]	0.014 [-0.122, 0.150]
FTM/S	0.37	1.09	-0.71	[-15.06, 27.73]	0.060 [-0.045, 0.164]
FIL	-7.86	1.06	-8.92	[-18.86, 4.24]	-0.038 [-0.158, 0.081]
VET	3.37	0.90	2.47	[-7.69, 24.73]	0.089 [-0.038, 0.216]

B01 | basket 2323 | January-September 2026

Buffered momentum; 28.0d lookback, 24h rebalance. **Net period return: -18.98%.** The eight net contributions below sum to that unchanged return.

Joint fit **273/273 days**; R-squared **0.238**; standardized condition **8.78**; max VIF **8.95**. Neither declared collinearity warning threshold was crossed. Max leverage **0.302**, max Cook's distance **1.829**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.046 [-0.108, 0.016]**; matched-calendar HAC27 **-0.046 [-0.112, 0.021]**. All brackets are marginal 95% intervals.

B01: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	2.21	1.68	0.53	[-6.11, 6.86]	-0.036 [-0.218, 0.145]
ETH	14.96	1.41	13.56	[5.78, 23.54]	0.021 [-0.100, 0.142]
BNB	-6.24	1.61	-7.85	[-20.96, 3.35]	0.134 [0.015, 0.253]
XRP	-12.57	1.69	-14.26	[-24.93, -6.59]	-0.036 [-0.153, 0.082]
DOGE	-6.58	1.93	-8.51	[-22.52, 3.14]	-0.102 [-0.264, 0.059]
FTM/S	5.42	2.17	3.25	[-24.68, 38.00]	-0.077 [-0.177, 0.023]
AXS	-8.89	1.48	-10.38	[-37.36, 28.84]	0.107 [-0.037, 0.250]
VET	6.16	1.48	4.68	[-7.79, 24.02]	0.012 [-0.124, 0.147]

B03 | basket 2457 | January-September 2026

Momentum; 90.0d lookback, 72h rebalance. **Net period return: -45.36%.** The eight net contributions below sum to that unchanged return.

Joint fit **273/273 days**; R-squared **0.290**; standardized condition **8.36**; max VIF **8.99**. Neither declared collinearity warning threshold was crossed. Max leverage **0.331**, max Cook's distance **1.213**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.113 [-0.193, -0.033]**; matched-calendar HAC27 **-0.113 [-0.195, -0.031]**. All brackets are marginal 95% intervals.

B03: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	-2.41	0.46	-2.87	[-11.58, 4.16]	0.018 [-0.123, 0.158]
ETH	-0.84	0.57	-1.41	[-10.29, 9.14]	0.016 [-0.056, 0.089]
BNB	-2.56	0.55	-3.10	[-14.17, 7.61]	0.128 [0.026, 0.229]
XRP	-6.74	0.66	-7.40	[-14.19, 0.57]	0.015 [-0.072, 0.102]
FTM/S	-1.10	0.34	-1.45	[-26.71, 25.11]	-0.189 [-0.244, -0.134]
MATIC/POL	-7.40	0.77	-8.18	[-17.00, 2.15]	-0.008 [-0.097, 0.081]
AXS	-25.35	0.45	-25.80	[-45.27, -5.56]	-0.004 [-0.110, 0.102]
LTC	5.51	0.66	4.85	[0.12, 11.44]	-0.032 [-0.115, 0.051]

B04 | basket 6064 | January-September 2026

Reversal; 90.0d lookback, 168h rebalance. **Net period return: -14.29%.** The eight net contributions below sum to that unchanged return.

Joint fit **273/273 days**; R-squared **0.735**; standardized condition **7.91**; max VIF **6.62**. Neither declared collinearity warning threshold was crossed. Max leverage **0.410**, max Cook's distance **2.681**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **0.026 [-0.034, 0.086]**; matched-calendar HAC27 **0.026 [-0.046, 0.098]**. All brackets are marginal 95% intervals.

B04: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	-0.81	0.23	-1.03	[-12.31, 10.15]	-0.123 [-0.175, -0.072]
ETH	-3.76	0.30	-4.06	[-18.85, 5.22]	-0.048 [-0.125, 0.028]
DOT	1.75	0.36	1.40	[-14.20, 17.75]	0.103 [0.036, 0.171]
LUNA/LUNC	-10.46	0.16	-10.61	[-42.70, 22.46]	-0.193 [-0.226, -0.160]
AVAX	2.77	0.45	2.32	[-10.78, 22.58]	0.065 [0.022, 0.107]
MANA	-1.35	0.38	-1.73	[-13.89, 13.82]	0.119 [0.051, 0.187]
FIL	14.62	0.53	14.09	[2.95, 27.96]	0.057 [0.017, 0.097]
VET	-14.29	0.36	-14.65	[-32.85, -0.89]	0.013 [-0.105, 0.132]

B05 | basket 2330 | January-September 2026

Trend long/cash; 28.0d lookback, 168h rebalance. **Net period return: -5.86%.** The eight net contributions below sum to that unchanged return.

Joint fit **273/273 days**; R-squared **0.496**; standardized condition **9.13**; max VIF **8.87**. Neither declared collinearity warning threshold was crossed. Max leverage **0.300**, max Cook's distance **0.626**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **0.311 [0.138, 0.485]**; matched-calendar HAC27 **0.311 [0.105, 0.517]**. All brackets are marginal 95% intervals.

B05: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	-0.09	0.31	-0.40	[-3.64, 3.93]	-0.068 [-0.220, 0.084]
ETH	3.16	0.14	3.02	[-3.49, 14.53]	0.022 [-0.087, 0.132]
BNB	-0.41	0.22	-0.63	[-6.77, 7.85]	0.019 [-0.096, 0.133]
XRP	-3.65	0.27	-3.92	[-8.11, -0.40]	0.046 [-0.106, 0.197]
DOGE	-3.24	0.27	-3.51	[-9.23, 2.88]	0.012 [-0.081, 0.104]
TRX	1.73	0.22	1.51	[-2.35, 6.43]	0.119 [0.024, 0.214]
GALA	-3.86	0.23	-4.08	[-11.66, 4.34]	0.039 [-0.017, 0.095]
LINK	2.44	0.31	2.14	[-6.67, 17.12]	0.164 [0.057, 0.271]

B06 | basket 2495 | January-September 2026

Trend long/short; 28.0d lookback, 168h rebalance. **Net period return: -4.22%.** The eight net contributions below sum to that unchanged return.

Joint fit **273/273 days**; R-squared **0.220**; standardized condition **9.00**; max VIF **8.88**. Neither declared collinearity warning threshold was crossed. Max leverage **0.319**, max Cook's distance **0.962**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.370 [-0.711, -0.028]**; matched-calendar HAC27 **-0.370 [-0.763, 0.024]**. All brackets are marginal 95% intervals.

B06: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	-0.76	0.65	-1.42	[-10.48, 9.69]	-0.273 [-0.561, 0.015]
ETH	6.89	0.28	6.62	[-2.98, 20.01]	-0.093 [-0.322, 0.135]
BNB	-0.18	0.47	-0.65	[-7.20, 6.96]	-0.115 [-0.336, 0.105]
XRP	-7.38	0.56	-7.95	[-21.67, 4.42]	-0.058 [-0.355, 0.239]
TRX	1.26	0.48	0.78	[-3.78, 5.16]	0.104 [-0.108, 0.317]
MANA	-0.88	0.47	-1.35	[-10.63, 8.62]	-0.037 [-0.191, 0.116]
GALA	-0.73	0.47	-1.20	[-17.61, 16.62]	-0.038 [-0.149, 0.074]
LINK	1.60	0.66	0.94	[-9.97, 16.14]	0.211 [0.022, 0.399]

Full precision, sensitivity and multiplicity

Per-coin gross, fee and net attribution with all three confidence intervals | Joint alpha/betas, HAC7 sensitivity, global Holm243 p-values and Bonferroni243 intervals | Joint model diagnostics and individual VIFs | Full-calendar BTC-only estimates and intervals | Matched BTC-only coefficients and intervals | Regression calendar and exclusions.

Attribution uses **4,000 seeded, synchronized circular 28-day calendar-block draws** across all nine strategies and eight coins. Daily gross/fee vectors are resampled together and relinked using each draw's **new prior NAV**; signals, targets and trades are not regenerated and no block-boundary fee is invented. Any invalid draw makes the affected interval unavailable; none is silently dropped. Gross/fee intervals and invalid-draw counts remain in the CSV.

The coefficient correction spans **243 predeclared diagnostic slots: 27 joint alphas + 216 joint betas** across all three periods. Displayed intervals are marginal; approximate Bonferroni familywise intervals and Holm-adjusted p-values are exported. This corrects only that conditional coefficient family, **not the original universe, configuration, period or strategy-selection search**. Attribution intervals are marginal and not multiplicity-adjusted. No significance row changes the frozen shortlist or crowns a new winner. LUNA/LUNC is legacy Terra, not Terra 2.0; FTM/S and MATIC/POL retain the original economic slots.

Performance above a 3% nominal cash benchmark

January-September 2026 evaluation; all nine unchanged strategies. Cash accrues at c = 0.03/365 each UTC calendar day (ACT/365F), compounded daily: **2.27% over 273 days**; the 365-day effective return is 3.0453%. This is a hypothetical constant **USDT cash proxy**, not an actual yield product; a USD interpretation assumes USDT/USD parity. **No interest is credited to strategy cash or short collateral.**

Raw returns versus cash-relative performance; drawdown uses DAILY endpoints

ID	Raw net return	Return vs cash (wealth-relative)	Relative CAGR	Excess Sharpe [95% CI], 365 / 252	Relative daily Max DD
A01	22.24%	19.53%	26.93%	365 1.07 [-0.41, 2.42] 252 0.89 [-0.34, 2.01]	12.82%
A02	14.36%	11.82%	16.11%	365 0.70 [-1.04, 2.03] 252 0.59 [-0.87, 1.69]	14.36%
A03	31.38%	28.46%	39.77%	365 2.13 [0.64, 3.64] 252 1.77 [0.53, 3.03]	6.21%
A04	-16.93%	-18.77%	-24.27%	365 -0.28 [-2.77, 1.64] 252 -0.23 [-2.31, 1.36]	39.02%
B01	-18.98%	-20.78%	-26.76%	365 -0.92 [-3.60, 1.14] 252 -0.76 [-2.99, 0.95]	34.33%
B03	-45.36%	-46.57%	-56.75%	365 -2.75 [-5.24, -1.12] 252 -2.29 [-4.36, -0.93]	46.57%
B04	-14.29%	-16.19%	-21.04%	365 -0.86 [-3.03, 1.65] 252 -0.71 [-2.51, 1.37]	31.53%
B05	-5.86%	-7.95%	-10.48%	365 -0.33 [-2.67, 1.78] 252 -0.28 [-2.22, 1.48]	22.89%
B06	-4.22%	-6.35%	-8.39%	365 -0.03 [-1.60, 1.19] 252 -0.02 [-1.33, 0.99]	20.25%

Cash-excess alpha: percent per year, not CAGR; intervals are marginal

ID	BTC-only alpha/yr [95% CI], full HAC7	Matched BTC alpha/yr [95% CI], HAC27	Joint basket alpha/yr [95% CI], HAC27	Mean-excess HAC27 t
A01	27.17% [-25.94%, 80.28%]	27.17% [-11.11%, 65.45%]	6.38% [-31.90%, 44.66%]	1.40
A02	18.43% [-37.82%, 74.68%]	18.43% [-25.71%, 62.58%]	0.74% [-45.44%, 46.92%]	0.81
A03	34.79% [3.03%, 66.55%]	34.79% [10.92%, 58.66%]	37.40% [19.06%, 55.75%]	2.83
A04	-13.90% [-126.21%, 98.41%]	-13.90% [-116.52%, 88.73%]	-17.32% [-111.75%, 77.11%]	-0.25
B01	-26.85% [-98.94%, 45.24%]	-26.85% [-91.90%, 38.20%]	-42.68% [-78.47%, -6.88%]	-0.79
B03	-79.39% [-140.08%, -18.69%]	-79.39% [-128.00%, -30.78%]	-84.70% [-126.25%, -43.15%]	-3.17
B04	-20.75% [-75.29%, 33.79%]	-20.75% [-77.31%, 35.81%]	-3.07% [-26.05%, 19.90%]	-0.72
B05	-8.50% [-48.52%, 31.53%]	-8.50% [-39.27%, 22.28%]	-13.28% [-42.07%, 15.51%]	-0.30
B06	-0.59% [-74.40%, 73.22%]	-0.59% [-55.17%, 54.00%]	-16.66% [-69.97%, 36.64%]	-0.04

Definitions. Return vs cash = W(strategy)/W(cash) − 1, not the difference of cumulative returns or a compounded arithmetic excess. Relative CAGR annualizes this wealth ratio. Excess Sharpe uses daily r − c, sample standard deviation and $\sqrt{365}$ for the primary convention; the same statistic and CI endpoints are multiplied by $\sqrt{(252/365)}$ for the companion 252-day convention. No daily observations or weekends are removed. Relative daily drawdown includes initial NAV and is **not comparable to the original hourly Max DD** as evidence of improved risk.

Inference. Sharpe intervals use 4,000 synchronized circular 28-day calendar-block draws, seed 20261016; return/growth intervals are exported. Mean-excess t/95% intervals use calendar HAC27 and n/(n−1); a **new Holm27 family** spans nine strategies × three periods. Factor regressions subtract c from both strategy and all coin returns, retaining their original sample: joint/matched BTC use HAC27 with n/(n−k); full-calendar BTC uses HAC7 without correction. Alpha intervals are recalculated; betas and beta intervals are unchanged. A **separate new cash-coefficient Holm243/Bonferroni243 family** is exported; raw-return corrections remain intact.

Scope. These conditional diagnostics are not adjusted for the original basket/configuration/selection search. Frozen returns, trades, turnover, selections, raw-return risk limits and acceptance flags remain unchanged. No financing model or new winner is introduced.

Full precision, all three periods: Performance and bootstrap intervals | Combined summary | Mean tests and Holm27 | Joint coefficients and cash Holm243/Bonferroni243 | Matched BTC | Full-calendar BTC | Model samples.